SAI ABHINAV

Aspiring AI / ML Engineer

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SUMMARY

MCA student and applied ML engineer who builds and deploys end-to-end systems across reinforcement learning, computer vision, and LLM workflows. Shipped a live, production-deployed RL traffic-optimization app (PPO + SUMO) that cut vehicle wait time ~28%, plus CNN detection models reaching 92-94% accuracy. Seeking a Junior AI/ML Engineer role to build production ML and GenAI systems.

TECHNICAL SKILLS

Languages: Python, SQL

ML / Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn; CNNs / Computer Vision; Reinforcement Learning (PPO, Stable-Baselines3); Feature Engineering; Model Optimization

GenAI / LLM: OpenAI API, Prompt Engineering, Structured-Output Workflows

Deployment & Tools: Git & GitHub, Flask (REST APIs), Render, SUMO

Data: Pandas, NumPy, Matplotlib, Seaborn, Excel / Google Sheets

PROJECTS

SmartSignal - RL Traffic-Signal Optimization

Live Demo Code

Python - PPO (Stable-Baselines3) - SUMO - TomTom API - Flask - Render

- Built and deployed to production a reinforcement-learning agent (PPO) controlling multi-junction traffic signals in SUMO, cutting average vehicle wait time ~28% and raising throughput 22% vs. fixed-timer baselines.
- Integrated the live TomTom API to drive simulations from real-world traffic data, and engineered the environment to scale across multiple junctions concurrently.
- Served the trained model behind a Flask REST app hosted on Render for live demonstration.

AI Workflow Copilot - LLM Document Automation

Code

Python - OpenAI API - Prompt Engineering - Flask

- Built an LLM tool that summarizes unstructured documents and auto-generates tasks and insights, using structured-output prompt workflows to return reliable, parseable JSON.
- Designed reusable prompt chains for extraction and task generation, reducing manual document-review effort.

Instagram Fake-Profile Detection

Code

Python - scikit-learn - TensorFlow/Keras (CNN) - Pandas

- Engineered an ensemble of classical ML + a CNN image classifier on a 3,000+ account dataset, reaching 92% accuracy, 90% precision, and 88% recall.
- Cut false positives 15% through feature engineering and decision-threshold tuning.

EXPERIENCE

Data Analyst Intern - IBM SkillsBuild — Jun 2024 - Aug 2024

Code: github.com/SaixAbhinav/skin_cancer_prediction

- Built and tuned a CNN for skin-cancer image classification, reaching 94% accuracy via data preprocessing, feature engineering, and hyperparameter tuning.
- Collaborated in a 6-person team and presented model findings to stakeholders.

EDUCATION

Master of Computer Applications (MCA) — 2025-2027 (Expected)

Vivekananda Institute of Professional Studies, Delhi | CGPA 8.6

Bachelor of Computer Applications (BCA) — 2022-2025

Vivekananda Institute of Professional Studies, Delhi | CGPA 7.6